



DSA in C++

Basics

Of C++

Lecture 03

Input in C++ // square of a number

take a number input

5

25

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Who is this batch for? → College students, working professionals
↓
B.Tech & BCA

Detailed tutorials on college wallah & camers channel for
VS Code & MinGW Installation & Setup.

↓
✓
32 bit
↓
Online Compiler
GDB

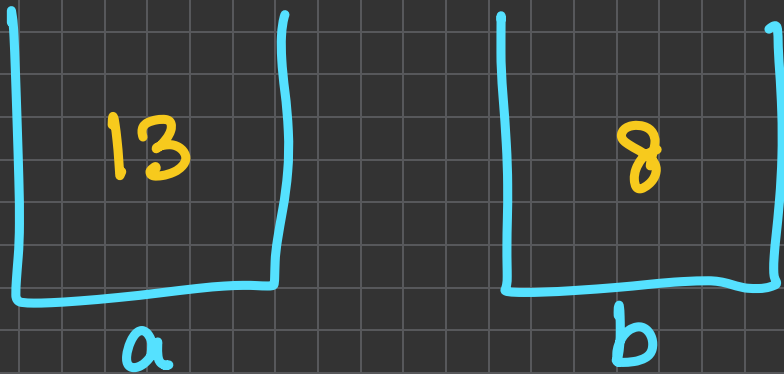


CW

Example: Take 2 numbers input and print their sum

5:03

```
✓int a,b;  
✓cout<<"Enter 1st number: ";  
✓cin>>a;  
✓cout<<"Enter 2nd number: ";  
✓cin>>b;  
✓cout<<a+b;
```



Output / Input

Enter 1st number: 13

Enter 2nd number: 8

21

Example: Calculate Simple Interest

$$SI = \frac{PRT}{100}$$

float p, r, t

Modulus Operator (remainder)

$+, -, *, /, \%$

$a \% b \rightarrow$ gives remainder
when a is divided by b

$$\begin{array}{r} 2 \\ 4 \overline{) 9} \\ \underline{8} \\ 1 \end{array}$$

Properties of Modulus Operator

$$2 \% 7 \rightarrow 2$$

$$3 \% 4 \rightarrow 3$$

$$7 \% (-2) \rightarrow 7 \% 2 = 1$$

$$a \% b = a \quad (\text{if } a < b)$$

$$a \% (-b) = a \% b$$

$$(-a) \% b = -(a \% b)$$

int/int
double/int

int/double
double/double

char Data Type

int , float

ASCII Values

'A' → 65

'B' → 66

'C' → 67

'D' → 68

⋮

⋮

'Y' → 89

'Z' → 90

'a' → 97

'b' → 98

'c' → 99

⋮

⋮

⋮

⋮

⋮

'z' → 122

'0' → 48

'1' → 49

'2' → 50

'3'

⋮

⋮

⋮

⋮

⋮

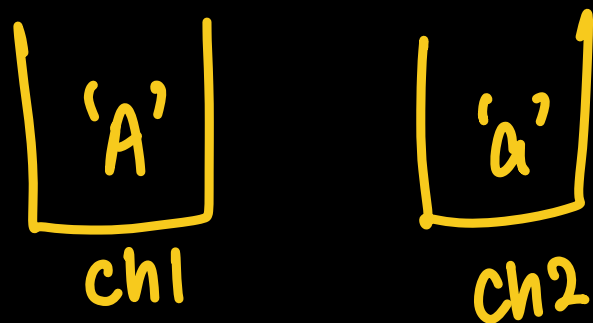
'9' → 57

Typecasting

int → char

char → int

```
✓ char ch1 = 'A';  
✓ char ch2 = 'a';  
✓ cout << ch1 + ch2 << endl;
```



char op char → int
↓
+, -, *, /

BODMAS (int)

left to right

Maths

$$x = 2 / 3 \times 6$$



$$2 / 3 \times 6$$

$$\frac{2}{3} \times 6$$

$$0.\overline{66} \times 6$$

4

$$\frac{12}{3} = 4$$

D > M > A > S

D, M > A, S

C++

$$x = 6 \times 2 / 3$$



$$6 \times 2 / 3$$

$$12 / 3$$

4

✓

$$6 \times 2 / 3$$

$$6 \times 0$$

0

✗

Post

$++x, x++, --x, x--$

post/pre increment/decrement operators

```
✓ int x = 10;  
✓ cout << x++ << endl;  
✓ cout << x << endl;
```

Output

10

11

pehle istemal karo
fir increment karo

11
10
x

Pre

$++x, x++, --x, x--$

post/pre increment/decrement operators

```
✓ int x = 10;  
✓ cout << ++x << endl;  
✓ cout << x << endl;
```

Output

11

11

pehle increment karo
fir print karo

11
10
x

++x, x++, --x, x--

✓ `int x = 8;`

✓ `int y = x++ + --x;`

`cout << y << endl`

8
9
8
x

y

x++ + --x
8 8

++x, x++, --x, x--

HW

```
int x = 8;
```

```
int y = x++ + ++x;
```

```
cout << y << endl
```

Mark True or False

1. Each new C++ instruction has to be written on a separate line *False*
2. Usually all C++ statements are entered in small case letters *True*
3. Blank spaces may be inserted b/w 2 words in a C++ Statement *True*